Representation Theory

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1 Group Theory

Group theory provides the algebraic language underlying representation theory. We recall the basic definitions and results used throughout these notes.

1.1 Definition of a Group

Definition 1.1 (A Group)**.** A Group consists of a set G together with a binary operation $\cdot$. This binary operation can be viewed as a rule for combining any 2 elements $g, h \in G$. Note, instead of writing $g \cdot h$, we will write gh instead. The set and binary operation must satisfy the following axioms:

1. for all $g, h, k \in G$:
$$(gh)k = g(hk)$$
2. there exists an element $e \in G$ such that for all $g \in G$:
$$eg = ge = g$$
3. for all $g \in G$, there exists $g^{-1} \in G$ such that:
$$gg^{-1} = g^{-1}g = e$$

axiom (1) is the associative property, the element e is the identity element of G and g^{-1} is known as the inverse of g

For any group G, there is only 1 identity element and every element g has at most 1 inverse. These 2 statements are relatively simple to prove and are left as an exercise for the reader. The product of an element g with itself gg is written as g^2. Similarly, $gg \dots g = g^n$ and $g^{-1} \dots g^{-1} = (g^{-1})^n$. Also $g^0 = e$. The number of elements in G is called the order of G and is written $|G|$.

The classic example of a group is $(\mathbb{Z}, +)$. An example of a finite group is called the cyclic group of order n: $C_n = \{e, a, a^2, \dots, a^{n-1}\}$.

A group G is said to be abelian if $gh = hg \ \forall g, h \in G$. $\mathbb{Z}$ and C_n are both abelian groups.

1.2 Subgroups

Definition 1.2 (Subgroup)**.** Let G be a group. A subset H of G is said to be a subgroup of G if it a group under the binary operation inherited from G. We use $H \leq G$ to indicate that H is a subgroup of G

The trivial group $\{e\}$ and the whole group G are always subgroups of G. Another classic subgroup of G is as follows:

Let G be a group and let $g \in G$. The subset:

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$$

Is a subgroup of G and is called the cyclic subgroup generated by g. If $g^n = e$ for some $n \geq 1$, then $\langle g \rangle$ is finite. You can generate other groups using more than 1 element from the group:

$$\langle g, h \rangle = \{g^n h^m : n, m \in \mathbb{Z}\}$$

A very important group with a large amount of subgroups is called the General Linear group, denoted $GL(n, K)$. This is the group of $n \times n$ invertible matrices over a field K. As we go further in the notes, we will use this group more and more.

1.3 Functions in group theory

Definition 1.3 (Group Homomorphisms)**.** If G and H are groups, then a homomorphism from G to H is a function $f : G \to H$ which satisfies:

$$f(gh) = f(g)f(h) \quad \forall g, h \in G$$

In other words, the function preserves the group operation. If the homomorphism is invertible, then we call it an isomorphism and in this case, we call G and H isomorphic and we write $G \cong H$ to denote that. An isomorphism means that G and H are essentially the same group.

From this definition, it clearly follows that if f is an isomorphism between G and H, then f^{-1} is also an isomorphism between H and G. It also follows that group properties such as commutativity, order and index are all the same between the 2. One last important properties of homomorphisms is that the identity of G will always be mapped to the identity of H.

Definition 1.4 (Images and Kernel)**.** Let G and H be groups and let $f : G \to H$ be a homomorphism. We define the image of f as:

$$\text{Img } f = \{f(g) : g \in G\}$$

And the kernel of f as:

$$\ker f = \{g \in G : f(g) = e_H\}$$

Where e_H represents the identity element of H

The image of f is a subgroup of H and the kernel of f is a subgroup of G. Furthermore, f is said to be injective if and only if the kernel of f is the trivial group $\{e\}$

1.4 Cosets

Definition 1.5 (Coset)**.** Let G be a group and let $H \leq G$. for $g \in G$, the subset;

$$Hg = \{hg : h \in H\}$$

Is said to the right coset of H in G.

All distinct right cosets of H in G form a partition of G which means that every element in G is in exactly one of the cosets.

If G is finite and H is a subgroup of G, then $|H|$ divides $|G|$. This is known as Lagrange's theorem and will become very useful later on. We can also write Lagrange's theorem as:

$$|G| = [G : H] \times |H|$$

Where $[G : H]$ is the number of right cosets of H in G

1.5 Normal subgroups

Definition 1.6 (Normal subgroups)**.** A subgroup N of G is said to be a normal subgroup of G if $g^{-1}Ng = N$ for all $g \in G$. We use $N \lhd G$ to indicate that N is a normal subgroup of G

Using G and N, we can create a new group known as the factor group G/N. This a group which consists of all the right cosets of N in G.

The trivial group and whole group are always normal subgroups of G. Furthermore, if we have a homomorphism f between G and some other group H, then the kernel of f is a normal subgroup of G.

Definition 1.7 (Simple Group)**.** A group G is said to be simple if $G \neq \{e\}$ and the only normal subgroups of G are itself and the trivial group.

If G is a finite group and not simple, then G is built up from other simple groups. In a way, simple groups play a role in group theory analogous to prime numbers in the integers.

These constructions are essential for studying group representations, particularly when considering homomorphisms into linear groups such as $GL(n, K)$.